

Predicting Passenger Anxiety with a Spatio-Temporal PIGNN in VR Autonomous Driving

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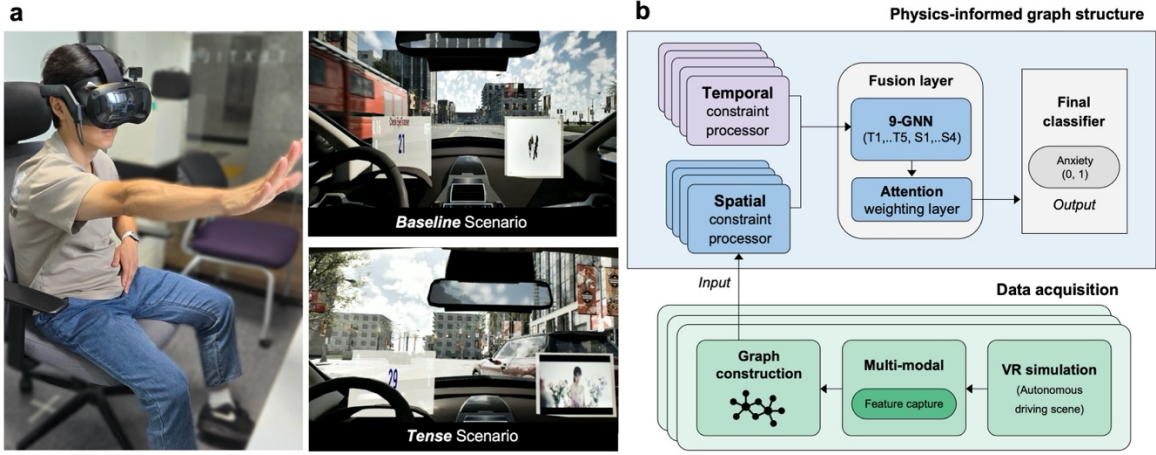


Figure 1: (a) Participant performing a VR-based autonomous driving task. (b) Overview of our pipeline: multimodal data from the VR simulation is processed into nine temporal and spatial constraint-based graphs. Each constraint is modeled by a dedicated GNN processor, and the outputs are fused via an attention-weighted layer to predict anxiety.

Abstract

Understanding passenger emotions across different autonomous driving modes is essential for designing adaptive in-vehicle systems. We present a spatio-temporal physics-informed graph neural network (PIGNN) to classify passenger anxiety in a virtual reality (VR) autonomous driving environment. Our model incorporates five temporal constraints (e.g., gaze velocity) and four spatial constraints (e.g., lateral motion), each processed through separate graph neural network (GNN) branches. Among the three model configurations,

the attention-weighted multi-constraint architecture achieved the highest accuracy (96.6%) and the lowest test loss (0.5) in classifying passenger anxiety across baseline and tense driving scenarios. By integrating domain-specific physics-informed constraints, our model captures temporally and spatially evolving emotional patterns within physically plausible boundaries, enabling more realistic classification of passenger anxiety.

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1 Introduction

As autonomous driving advances, understanding passenger emotions is crucial for ensuring safe and comfortable experiences [Kim

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et al., 2022]. Unlike traditional driver monitoring, passenger emotion recognition must address subtle behavioral cues and dynamic contexts [Palomares et al., 2022]. Although driving modes are often treated uniformly, passengers may exhibit distinct emotional responses based on situational dynamics and their perceived level of control. However, few studies have explored how these emotions unfold across varying spatio-temporal contexts.

In this study, we present a framework to detect passenger anxiety in VR-based autonomous driving, focusing on emotional changes triggered by risk events. Leveraging VR technology, we create a controlled yet immersive simulation environment that closely mirrors real-world conditions. Our system integrates behavioral and spatial data to model both the temporal dynamics and contextual interactions of passengers. We construct graphs grounded in domain-specific physical constraints and process them through a spatio-temporal PIGNN. This dual modeling of time and space enables interpretable and accurate classification of anxiety-related states (Fig. 1). This study contributes: (1) a PIGNN framework that models emotional states through physics-informed spatio-temporal representations of behavior and context, (2) a fully autonomous VR simulation platform for emotion recognition, and (3) a constraint design that models temporal and spatial factors for emotion inference in autonomous driving.

2 Our approach

2.1 Method

To analyze passenger anxiety in autonomous driving, we developed a VR environment using the CARLA simulator, HTC Vive Focus 3, and an integrated eye tracker. Data from 41 participants included gaze, pupil size, head movement, and information about surrounding objects across baseline and tense scenarios. The tense scenarios involved four distinct hazard events, such as sudden cut-in vehicles and unexpected pedestrian crossings. We constructed five temporal graphs (fixation, gaze velocity, gaze entropy, pupil size, and head rotation) and four spatial graphs (distance change, lateral offset, proximity entropy, and pedestrian spacing), each representing a physics-informed constraint and processed by a dedicated GNN. We compared three multi-processor configurations: (1) a simple two-branch model grouping temporal and spatial constraints, (2) a nine-processor model assigning one GNN per constraint with equal fusion weights, and (3) an attention-weighted version that learns each constraint's relative importance. A decoder predicts anxiety from the fused embedding [Wang et al., 2024], trained using classification loss and auxiliary constraint-consistency losses.

2.2 Experimental results

We evaluated our model across baseline and tense driving scenarios. The initial two-branch model, which separated temporal and spatial constraints into two processors, achieved a high accuracy of 97.4% but exhibited overfitting, with a prediction loss of 1049.5 on the test set. Replacing this model with nine independent constraint processors improved generalization, reducing the test loss to 55.2 while maintaining an accuracy of 95.4%. The attention-weighted version further enhanced performance, achieving the lowest test loss of 0.5 and an accuracy of 96.6%. By decomposing the input into nine physics-informed processors and applying adaptive weighting

based on constraint relevance, the model effectively captured spatio-temporal patterns associated with passenger anxiety in autonomous driving.

3 Conclusion and discussion

In this paper, we presented a spatio-temporal PIGNN model that integrates temporal behavioral dynamics and spatial driving context under physics-informed constraints to predict passenger anxiety in fully autonomous driving. Our results highlight its effectiveness in capturing realistic emotional patterns, with future work focused on validating scalability across broader scenarios and more diverse passenger populations.

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